Federated Averaging for Multi-Class Network Intrusion Classification: Theoretical Foundations, Empirical Evaluation, and Resilience to Label Poisoning on the CIC–UNSW-NB15 Dataset

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Abstract—Federated learning (FL) enables multiple institutions to jointly train machine learning models without exchanging raw data, a property that is especially desirable for network security analytics where flow telemetry may contain sensitive operational information. This paper presents a comprehensive theoretical and empirical study of Federated Averaging (FedAvg) applied to multi-class intrusion / traffic classification on the CIC–UNSW-NB15 augmented dataset. We formalize the FL objective, the FedAvg update rule, local empirical risk minimization with multilayer perceptrons (MLPs), and a single-client label-poisoning threat model. We then conduct a 64-configuration experimental sweep over client counts, global communication rounds, local epochs, and poison intensities. Empirically, longer training improves clean accuracy (best clean accuracy $\approx$ 0.924), while targeted poisoning is most harmful when a single client dominates training (worst accuracy $\approx$ 0.856). With multiple clients, FedAvg dilutes one poisoned shard and accuracy typically remains above 0.90. Confusion-matrix and one-vs-rest TP/FP/FN/TN analyses show that majority-class targeted poisoning systematically increases false positives for class 0 and reduces recall for several minority attack categories—an effect that can be understated by overall accuracy alone under severe class imbalance. We discuss communication cost, convergence intuition under IID shards, limitations of the simulation setting, and directions for robust aggregation. An interactive dashboard accompanies the study for transparent inspection of all runs.

Index Terms—Federated learning, Federated Averaging (FedAvg), intrusion detection systems, network traffic classification, data poisoning, label flipping, CIC–UNSW-NB15, multilayer perceptron, confusion matrix, class imbalance

I. Introduction

A. Motivation

Network intrusion detection and traffic classification increasingly rely on supervised learning over high-dimensional flow features. In operational settings, however, the organizations that collect such telemetry—enterprises, internet service providers, campus networks, or industrial control environments—are often unwilling or legally unable to centralize raw records. Privacy regulations, competitive sensitivity, and breach-risk all discourage naive data pooling.

Federated learning [1], [2] offers an alternative: each participant trains on local data and shares only model parameters (or gradients), which a coordinating server aggregates into a global model. For intrusion detection, this paradigm suggests a path toward collaborative detection quality without collaborative data warehouses [3].

B. Problem Statement

We study the following concrete problem. Given a tabular multi-class dataset with feature vectors $\mathbf{x} \in \mathbb{R}^d$ and labels $y \in \{0, \ldots, C-1\}$, simulate N clients holding disjoint training shards and learn a shared classifier f_θ via FedAvg. We then ask:

1) How do N, the number of global rounds R, and local epochs E affect test accuracy and F1-score?
2) How does label poisoning on a single client, with fraction f of labels corrupted, change global performance?
3) Does poisoning alter which classes are confused, beyond aggregate metrics?

C. Scope and Simulation Nature

All experiments in this repository run on a single machine. "Clients" are logical partitions of a shuffled training set rather than physically distributed hosts. The simulation preserves the algorithmic structure of FL (local optimization + server aggregation) while abstracting away network latency, dropouts, and systems engineering. This is appropriate for controlled scientific comparison of learning dynamics and poisoning effects, but it is not a full systems deployment study.

D. Contributions

1) A self-contained theoretical presentation of FedAvg for multi-class classification, including the global objective, local surrogate losses, weighted aggregation, and a formal label-poisoning adversary.

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2) An end-to-end reproducible pipeline on CIC–UNSW-NB15 augmented data: cleaning, stratified splitting, scaling, IID sharding, MLP training, and evaluation with confusion matrices and per-class TP/FP/FN/TN.
3) A 64-run experimental grid quantifying interactions among clients, rounds, local epochs, and poison settings, with best-case clean accuracy near 0.924 and worst-case poisoned accuracy near 0.856.
4) Qualitative and quantitative evidence that majority-class targeted poisoning induces a "predict class 0 when uncertain" bias that is most severe under single-client training and largely diluted under multi-client FedAvg.
5) Public interactive visualizations of the full experimental record.

E. Paper Organization

Section II reviews FL, IDS learning, and poisoning. Section III develops the mathematical framework. Section IV describes the system design. Section V details the experimental protocol. Sections VI–VII present and interpret results. Section VIII concludes.

II. Background and Related Work

A. Centralized Learning for Intrusion Detection

Classical machine learning for IDS ranges from shallow models (decision trees, SVMs, random forests) to deep neural networks trained on packet or flow features. Centralized training assumes a single data lake. While often accurate in benchmarks, this assumption conflicts with multi-organization collaboration. Moreover, network data are characteristically imbalanced: benign or majority classes dominate, so accuracy can be misleading unless accompanied by per-class analysis.

B. Federated Learning Basics

In cross-silo or cross-device FL, a server coordinates rounds of local training and aggregation [2]. FedAvg [1] is the canonical algorithm: clients run multiple local SGD/Adam steps and return updated weights; the server averages them, typically weighted by local dataset sizes. Variants address non-IID data (FedProx, SCAFFOLD), communication compression, personalization, and robust aggregation.

C. FL for Security Analytics

Prior work has explored FL for IoT and network intrusion detection to keep telemetry local while improving collective detectors [3]. Our contribution is not a new FL optimizer; rather, we provide a carefully instrumented FedAvg case study with poisoning analysis and confusion-level diagnostics on a widely used tabular intrusion dataset.

D. Poisoning and Byzantine Behavior

Data poisoning injects corrupted training examples to degrade or redirect a learned model. In FL, a malicious client can poison labels or features, or upload crafted weights [4], [5]. Defenses include robust aggregators (coordinate-wise median, trimmed mean, Krum), anomaly detection on updates, and differential privacy. This paper evaluates undefended FedAvg against a simple but realistic label-flipping adversary on one client, establishing an empirical baseline for how much "natural dilution" multi-client averaging provides.

E. UNSW-NB15 Family of Datasets

UNSW-NB15 and related CIC-augmented releases provide labeled flow features covering multiple attack categories alongside normal traffic. We use an augmented CIC–UNSW-NB15 tabular release with $d = 76$ numeric features and $C = 10$ classes. Exact semantic names of minority classes are less critical for our algorithmic conclusions than the empirical imbalance pattern: one dominant class and several rare classes with near-zero recall under a simple MLP.

III. Theoretical Framework

A. Supervised Multi-Class Risk

Let $\mathcal{D}$ be a distribution over $\mathcal{X} \times \mathcal{Y}$ with $\mathcal{Y} = \{0, \ldots, C-1\}$. A parametric model $f_\theta : \mathcal{X} \to \mathbb{R}^C$ produces logits; predicted label is $\hat{y} = \arg\max_c [f_\theta(\mathbf{x})]_c$. The population risk under cross-entropy is

$$R(\theta) = \mathbb{E}_{(\mathbf{x},y)\sim\mathcal{D}}\Big[\ell\big(f_\theta(\mathbf{x}), y\big)\Big], \tag{1}$$

where $\ell(\mathbf{z}, y) = -\log \frac{e^{z_y}}{\sum_c e^{z_c}}$.

Given a finite training set $S = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$, the empirical risk is

$$\hat{R}_S(\theta) = \frac{1}{n}\sum_{i=1}^{n} \ell\big(f_\theta(\mathbf{x}_i), y_i\big). \tag{2}$$

B. Federated Partition of the Empirical Risk

Partition S into N disjoint client datasets S_k with $n_k = |S_k|$ and $\sum_k n_k = n$. The global empirical risk decomposes exactly as a mixture of local empirical risks:

$$\hat{R}_S(\theta) = \sum_{k=1}^{N} \frac{n_k}{n}\, \hat{R}_{S_k}(\theta). \tag{3}$$

Equation (3) is the conceptual justification for size-weighted FedAvg: averaging models trained on local risks approximates minimizing a weighted combination of those risks when local updates are not too far from the broadcast initialization.

C. Local Optimization

In round t, client k receives θ^t and approximately solves

$$\theta_k^{t+1} \approx \arg\min_{\theta} \hat{R}_{S_k}(\theta) \quad \text{initialized at } \theta^t, \tag{4}$$

using E epochs of minibatch Adam with learning rate η. Let B be the minibatch size. The number of local gradient steps per round is roughly $E \cdot \lceil n_k/B \rceil$. More local steps increase client drift away from θ^t but also increase local progress per communication round—the classic FedAvg trade-off.

D. FedAvg Aggregation

The server sets

$$\theta^{t+1} = \sum_{k=1}^{N} w_k \, \theta_k^{t+1}, \qquad w_k = \frac{n_k}{\sum_{j=1}^{N} n_j}. \tag{5}$$

If shards are equal-sized, $w_k = 1/N$. Aggregation is performed element-wise over all tensors in the state dictionary (weights and biases of every layer).

E. Communication Cost Model

Let $|\theta|$ denote the number of scalar parameters. Each round transmits approximately $N|\theta|$ parameters uplink and $N|\theta|$ downlink (broadcast counted per client), i.e., $\Theta(N|\theta|)$ communication per round. Total communication over R rounds scales as $\Theta(RN|\theta|)$. Increasing E reduces required R for a target accuracy in many IID settings, improving communication efficiency—one of FedAvg's original motivations [1].

F. MLP Hypothesis Class

Our model is a two-hidden-layer MLP:

$$f_\theta(\mathbf{x}) = W_3 \, \sigma\big(W_2 \, \sigma(W_1 \mathbf{x} + b_1) + b_2\big) + b_3, \tag{6}$$

with $\sigma = \text{ReLU}$, hidden width 128, $d = 76$, and $C = 10$. The hypothesis class is rich enough for nonlinear decision boundaries on tabular features, yet small enough for rapid experimentation. Softmax is implicit in the cross-entropy loss.

G. Class Imbalance and Metrics

Let $n^{(c)}$ be the number of test examples of class c. Overall accuracy is

$$\text{Acc} = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} \mathbf{1}[\hat{y}_i = y_i]. \tag{7}$$

Under imbalance, Acc is dominated by majority-class correctness. We therefore also report precision, recall, F1 (from scikit-learn's classification report aggregates), the full confusion matrix $M \in \mathbb{N}^{C \times C}$ with

$$M_{ij} = \#\{i' : y_{i'} = i, \ \hat{y}_{i'} = j\}, \tag{8}$$

and one-vs-rest counts

$$\text{TP}_c = M_{cc}, \tag{9}$$

$$\text{FP}_c = \sum_{i \neq c} M_{ic}, \tag{10}$$

$$\text{FN}_c = \sum_{j \neq c} M_{cj}, \tag{11}$$

$$\text{TN}_c = n_{\text{test}} - \text{TP}_c - \text{FP}_c - \text{FN}_c. \tag{12}$$

Per-class recall $\text{TP}_c/(\text{TP}_c + \text{FN}_c)$ is especially informative for rare attacks.

H. Threat Model: Label Poisoning

We consider a dirty-label adversary controlling client 0 only. Features remain untouched; a fraction $f \in [0, 1]$ of local labels are replaced according to a mode m:

- Target: $y \leftarrow y_{\text{tgt}}$ (we use $y_{\text{tgt}} = 0$).
- Random-wrong: $y \leftarrow$ uniform over $\mathcal{Y} \setminus \{y_{\text{true}}\}$.
- Cycle: $y \leftarrow (y + 1) \bmod C$.

The adversary does not modify uploaded weights beyond ordinary training on poisoned labels, and does not control the test distribution. This models a compromised or negligent data pipeline at one silo rather than an optimal model-replacement attack [4].

I. Why Dilution Occurs under FedAvg

Suppose equal shard sizes and one poisoned client. Even if θ_0^{t+1} is strongly biased, the aggregate is

$$\theta^{t+1} = \frac{1}{N}\theta_0^{t+1} + \frac{1}{N}\sum_{k=1}^{N-1} \theta_k^{t+1}. \tag{13}$$

As N grows, the poisoned update's coefficient $1/N$ shrinks. This is not a cryptographic defense and fails if many clients collude or if the adversary mounts stronger model poisoning; but it explains our empirical observation that multi-client runs remain near clean accuracy under single-client label flips.

J. Convergence Intuition (IID Case)

When shards are IID draws from the same distribution, local risks $\hat{R}_{S_k}$ are statistically similar. Local Adam steps from a shared initialization tend to move in related directions, so averaging is constructive. Formal FedAvg convergence bounds typically assume smoothness, bounded variance, and control of client drift as a function of E and heterogeneity [2]. Our shards are IID-like by construction (random shuffle then cut), placing us near the favorable regime. Non-IID partitions would increase drift and could amplify poisoning or reduce clean accuracy; we leave that to future work.

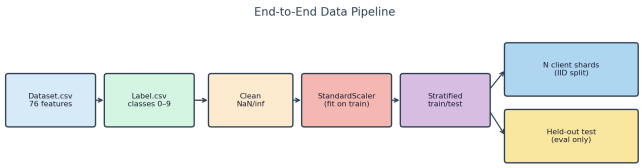


Fig. 1. End-to-end data pipeline from raw CSVs to client shards and a held-out test set.

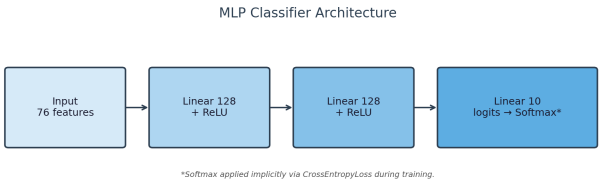


Fig. 2. Shared MLP classifier used by every client.

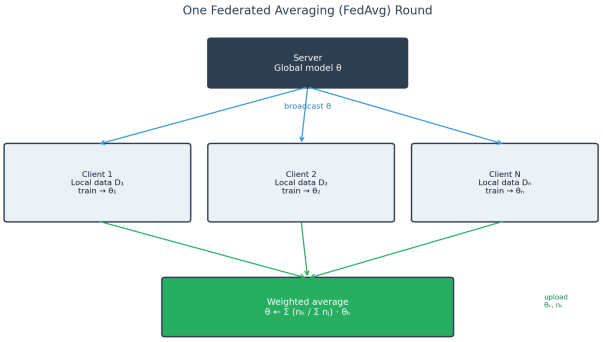


Fig. 3. One FedAvg round: broadcast, local training, and weighted aggregation.

IV. System Design and Implementation

A. Data Pipeline

Figure 1 summarizes preprocessing. Features and labels are aligned row-wise. Non-finite values are replaced by zeros. A stratified split (default 80% train / 20% test) preserves class proportions. A StandardScaler is fit on training features only, then applied to train and test. Training indices are shuffled with a fixed seed and split into N contiguous shards after permutation.

a) Note on global scaling.: Fitting a scaler on pooled training features is a practical simplification for a single-machine simulation. Strictly decentralized FL might estimate normalization statistics via secure aggregation or use per-client normalization. We explicitly document this assumption.

B. Model Architecture

Figure 2 shows the shared MLP. Identical architecture across clients is mandatory for element-wise FedAvg.

C. Federated Training Loop

Figure 3 and Algorithm 1 describe one round of training.

Algorithm 1 FedAvg for intrusion classification (simulation)

Require: clients $\{S_k\}_{k=1}^{N}$, rounds R, local epochs E, batch size B, learning rate η

initialize global model θ^0
for $t = 0$ to $R-1$ **do**
  for each client $k = 1, \ldots, N$ in parallel **do**
    $\theta_k \leftarrow \theta^t$
    for $e = 1$ to E **do**
      for each minibatch $\mathcal{B} \subset S_k$ **do**
        Adam update on $\hat{R}_{\mathcal{B}}(\theta_k)$ with step η
      end for
    end for
    upload (θ_k, n_k)
  end for
  $\theta^{t+1} \leftarrow \sum_k (n_k / \sum_j n_j)\, \theta_k$
  evaluate θ^{t+1} on held-out test set
end for
return θ^R

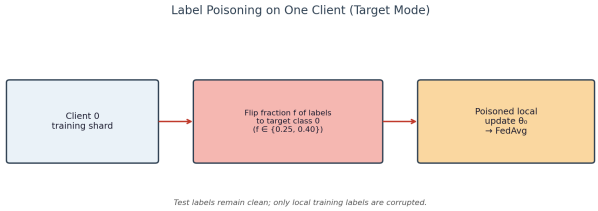


Fig. 4. Targeted label poisoning applied only to client 0 training labels.

D. Poisoning Hook

Before local training begins, if poisoning is enabled, client 0's label vector is rewritten according to Section III-H (Fig. 4). Features and all other clients remain clean.

E. Software Stack

The implementation uses PyTorch for the MLP and optimization, pandas/NumPy for data loading, and scikit-learn for splitting, scaling, confusion matrices, and classification reports. Experiments are driven by a configuration dataclass and CLI flags for clients, rounds, local epochs, batch size, learning rate, device, and poison parameters.

V. Experimental Setup

A. Hardware and Reproducibility

Runs use CPU by default with fixed random seeds for NumPy and PyTorch. This favors reproducibility over wall-clock speed.

B. Grid Design

The principal sweep comprises 64 configurations exploring:

- $N \in \{1, 2, 3, 5\}$,

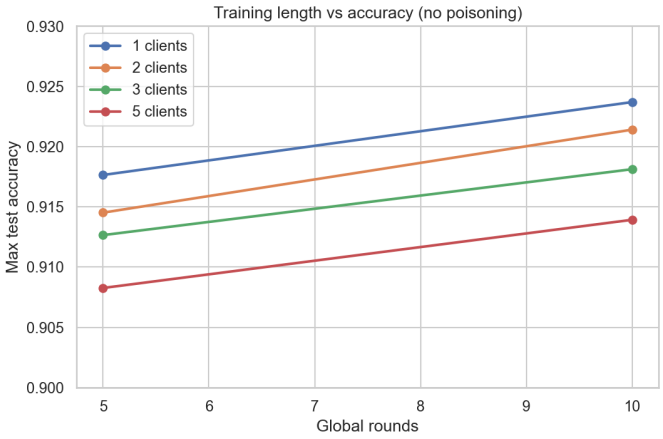


Fig. 5. Effect of global rounds on clean-run accuracy.

- global rounds (notably 5 and 10 in the grid),
- local epochs $E \in \{1, 2\}$,
- poison fraction $f \in \{0, 0.25, 0.40\}$,
- poison mode $\in \{\text{target}, \text{random_wrong}\}$ when $f > 0$.

Primary metrics: test accuracy, precision, recall, F1. Selected representative scenarios additionally store full confusion matrices.

C. Descriptive Statistics of the Sweep

Across all 64 runs, accuracy has mean ≈ 0.907, standard deviation ≈ 0.014, minimum ≈ 0.856, and maximum ≈ 0.924. The relatively small standard deviation indicates that most configurations remain in a high-accuracy band; the lower tail is concentrated in heavily poisoned single-client settings.

D. Evaluation Protocol

The test set is never poisoned and never used for training. After the final round (and optionally intermediate rounds), we compute aggregate metrics and, for key scenarios, confusion matrices and TP/FP/FN/TN.

E. Artifacts

Results are exported to CSV; static figures and an interactive HTML dashboard summarize the sweep. The live dashboard is available at https://fl-research.vercel.app.

VI. Results

A. Clean Training Behavior

Under clean labels, increasing R and E generally improves accuracy. Single-client clean training can slightly outperform multi-client FedAvg at the same budget because there is no averaging-induced damping of local specialization; however, multi-client accuracy remains strong (typically > 0.90). Figures 5 and 6 illustrate round and local-epoch effects.

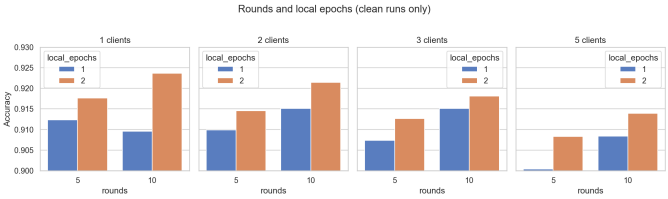


Fig. 6. Local-epoch effects for clean federated runs.

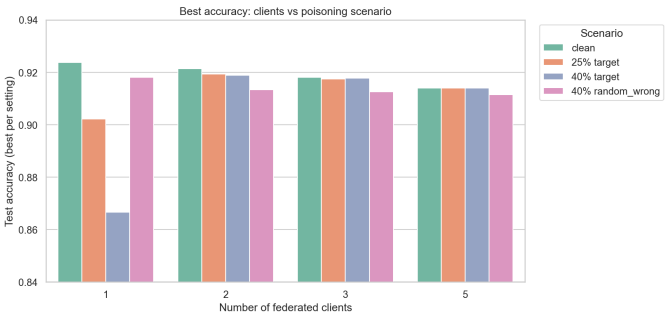


Fig. 7. Best accuracy by client count and poisoning scenario.

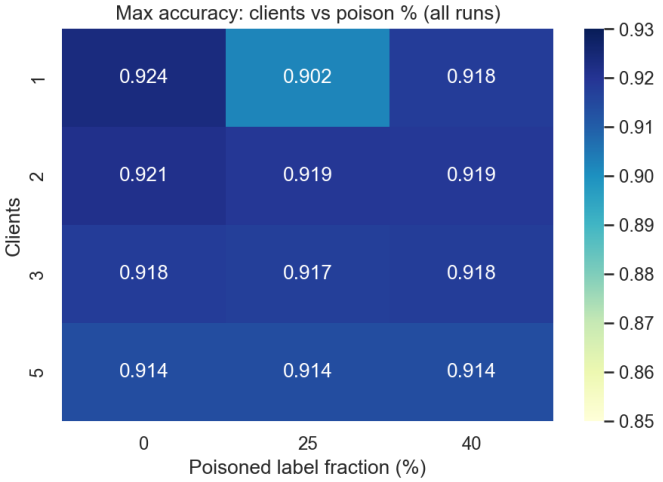


Fig. 8. Max accuracy heatmap over clients versus poison percentage.

B. Clients and Poison Scenarios

Figure 7 compares best accuracy by client count and scenario. Figure 8 shows a clients-versus-poison heatmap of maximum accuracy. Figure 9 contrasts clean versus 40% target poisoning.

C. Best Configurations

Table I lists best observed metrics by condition. The most damaging cell is $N = 1$ with 40% target poison (accuracy ≈ 0.867 in the best-of-condition table; some individual runs fall near 0.856). By contrast, $N = 5$ with 40% target poison remains essentially tied with the clean five-client best (≈ 0.914).

D. Mean Accuracy by Clients and Poison Fraction

Table II reports mean accuracy aggregated over rounds/epochs/modes in the sweep. The drop from $f = 0$

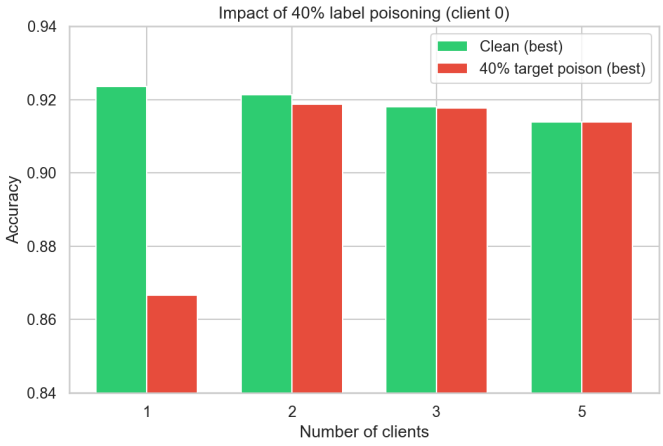


Fig. 9. Clean versus 40% target poisoning (best accuracy per client count).

TABLE I
Best test accuracy by client count and poison setting

Clients	Poison	Mode	Acc.	F1
1	0%	–	0.9237	0.9212
1	25%	target	0.9022	0.8881
1	40%	target	0.8666	0.8329
1	40%	random	0.9180	0.9127
2	0%	–	0.9214	0.9177
2	40%	target	0.9188	0.9132
3	0%	–	0.9181	0.9152
3	40%	target	0.9178	0.9135
5	0%	–	0.9139	0.9119
5	40%	target	0.9139	0.9117

TABLE II
Mean test accuracy by clients and poison fraction (sweep aggregate)

Clients	$f = 0$	$f = 0.25$	$f = 0.40$
1	0.9158	0.8981	0.8873
2	0.9152	0.9141	0.9106
3	0.9133	0.9127	0.9082
5	0.9077	0.9076	0.9051

to $f = 0.4$ is large for $N = 1$ ($\approx 0.916 \to 0.887$ mean) and small for $N = 5$ ($\approx 0.908 \to 0.905$).

E. Target versus Random-Wrong Poisoning

Target poisoning toward class 0 is typically more harmful than random-wrong poisoning at the same fraction in low-N regimes. Random-wrong noise is less coherently biased; target poisoning consistently teaches a majority-class prior. This distinction appears both in aggregate metrics and in confusion structure.

F. Confusion Matrices: Clean Multi-Client

Figures 10–11 show five-client clean versus poisoned matrices. They are nearly alike: FedAvg dilution keeps decision boundaries stable. Class 0 dominates correct predictions; rare classes remain difficult; class 5 absorbs many off-diagonal errors.

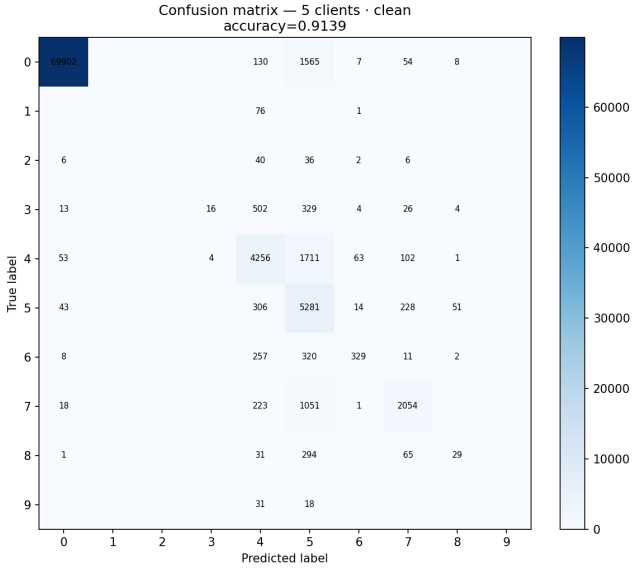


Fig. 10. Confusion matrix: 5 clients, clean training.

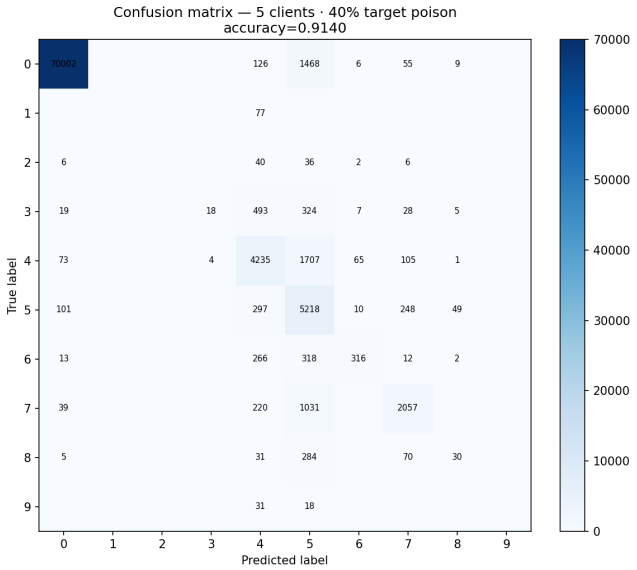


Fig. 11. Confusion matrix: 5 clients, 40% target poison on client 0.

G. Confusion Matrices: Single-Client Fragility

Figures 12–13 show the single-client contrast. After 40% target poisoning, substantial mass moves into predicted class 0 from true classes 4, 5, 6, and 7. Class 0 recall can remain high or even increase, while other classes suffer. Figures 14–15 make the same point via TP/FP/FN/TN bars: class 0 FP rises; FN rises for several non-zero classes.

H. Security Interpretation of Errors

From a SOC analyst's perspective, elevated class 0 false positives after target poisoning mean attack flows are more often declared "majority/benign-like" (depending on label semantics), i.e., missed detections. Overall accuracy may still look acceptable because class 0 is frequent. This is

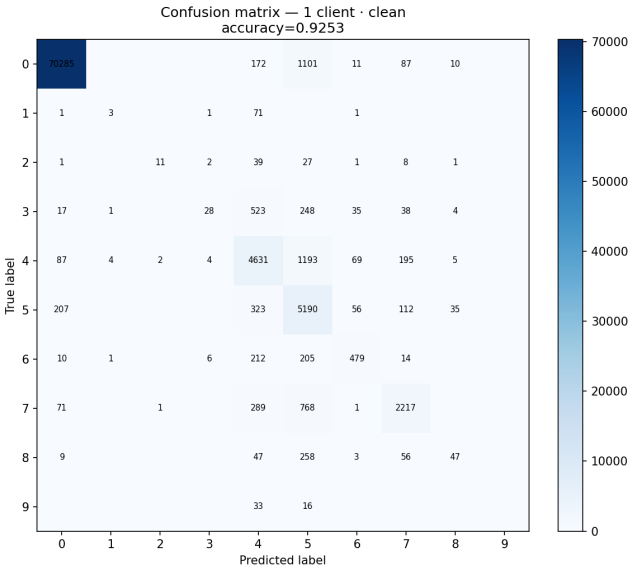


Fig. 12. Confusion matrix: 1 client, clean training.

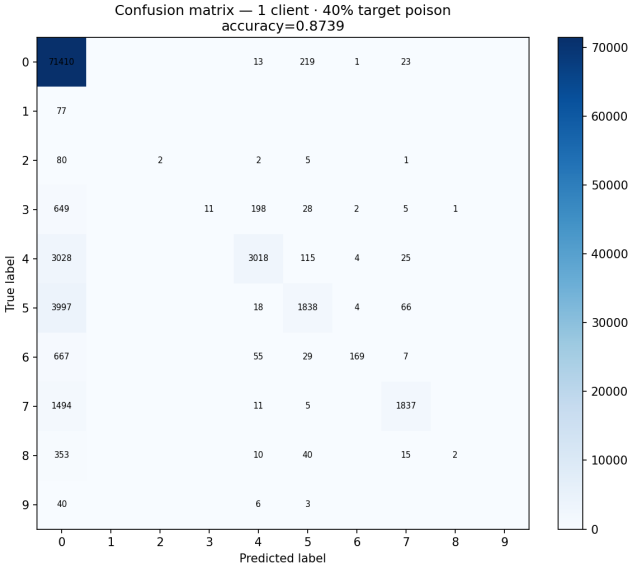


Fig. 13. Confusion matrix: 1 client, 40% target poison.

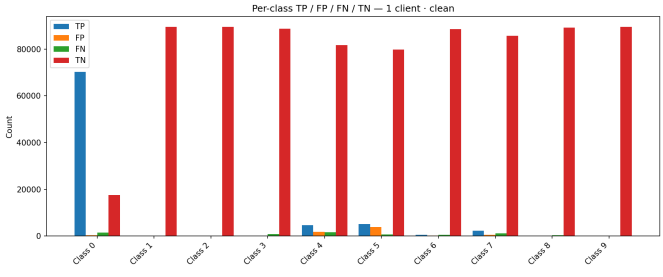


Fig. 14. Per-class TP/FP/FN/TN: 1 client, clean.

why confusion-level reporting is mandatory for security ML evaluations.

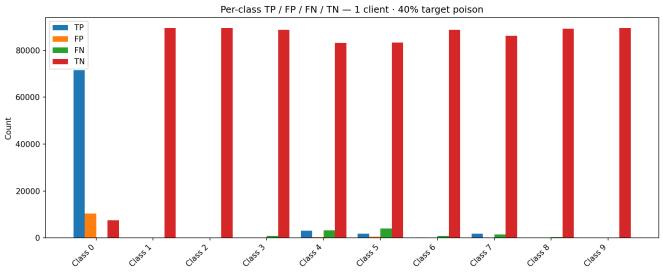


Fig. 15. Per-class TP/FP/FN/TN: 1 client, 40% target poison.

VII. Discussion

A. Implications for Deploying FL in IDS

1) Prefer multiple independent data silos when possible; averaging provides baseline robustness to one bad silo under label noise.
2) Prefer adequate R and E; under-trained models are both less accurate and potentially less stable under noise.
3) Monitor per-class recall, not only accuracy, especially for rare attack types.
4) Treat single-client "federated" training as essentially centralized: it inherits no dilution benefit.

B. What This Study Does Not Claim

We do not claim FedAvg is secure against adaptive model-poisoning or colluding clients. We do not claim MLP is the optimal IDS architecture. We do not claim IID shards represent real institutional heterogeneity.

C. Limitations

- IID simulation: random shards underestimate heterogeneity-induced drift.
- Global scaler: pooled normalization is a centralized shortcut.
- Single architecture: no comparison to trees, CNNs on raw packets, or GNNs.
- Undefended aggregation: no median/Krum/DP baselines.
- Label semantics: class indices are used without exhaustive attack taxonomy narration.

D. Threats to Validity

Random seeds, hyperparameter choices (Adam lr, batch size), and the particular augmented dataset release could shift absolute numbers. Relative conclusions—dilution with N, severity of target poison at $N = 1$, need for confusion analysis—were consistent across the sweep's internal variation.

E. Future Work

1) Non-IID label and feature skew across clients.
2) Robust aggregators and anomaly detection on updates.

3) Differentially private FL for formal privacy guarantees.
4) Cost-sensitive losses or resampling for rare classes.
5) Multi-poisoner and model-replacement attacks.
6) Deployment in a real FL framework with networked workers.

VIII. Conclusion

We presented an expanded theoretical and experimental study of FedAvg for multi-class traffic classification on CIC–UNSW-NB15. Starting from the global empirical risk and its client-wise decomposition, we formalized local optimization, size-weighted aggregation, communication scaling, class-imbalance metrics, and a single-client label-poisoning adversary. Empirically, among 64 configurations, clean training benefits from more rounds and local epochs; targeted poisoning is highly damaging when $N = 1$ and largely diluted when $N \geq 2$; and confusion matrices reveal majority-class bias that accuracy can mask. These findings support a practical message: federated collaboration with multiple honest clients, sufficient training, and per-class monitoring forms a stronger baseline than accuracy-only evaluation of single-silo training under label noise.

Acknowledgment

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Appendix A
Extended Notation and Optimization View

For completeness we collect notation used throughout the paper. Let N be the number of clients, R the number of global rounds, E the number of local epochs, B the minibatch size, and η the Adam learning rate. Client k holds n_k labeled examples. The global parameter vector $\theta \in \mathbb{R}^p$ stacks all MLP weights and biases. Softmax probabilities for class c are

$$p_c(\mathbf{x}; \theta) = \frac{\exp([f_\theta(\mathbf{x})]_c)}{\sum_{c'=0}^{C-1} \exp([f_\theta(\mathbf{x})]_{c'})}. \tag{14}$$

Cross-entropy on an example is $-\log p_y(\mathbf{x}; \theta)$. Adam updates use biased first/second moment estimates with standard (β_1, β_2) defaults in PyTorch; because we reinitialize Adam each round, moment state does not accumulate across communication rounds.

A useful optimization perspective treats FedAvg as repeatedly applying a local operator $\mathcal{U}_{k,E}$ ("E epochs of Adam on S_k") followed by consensus:

$$\theta^{t+1} = \sum_{k=1}^{N} w_k \, \mathcal{U}_{k,E}(\theta^t). \tag{15}$$

When $E = 1$ and minibatches equal full local sets, this resembles a single gradient step mixture method. Large E increases the nonlinearity of $\mathcal{U}_{k,E}$ and therefore the benefit (or harm) of averaging, depending on whether local trajectories diverge.

Appendix B
Additional Discussion of Metrics under Imbalance

Macro-averaged metrics treat each class equally, whereas micro-averaged / accuracy-weighted metrics emphasize frequent classes. In intrusion detection, both views matter: macro recall highlights rare-attack coverage, while micro accuracy reflects bulk traffic triage. Our sweep exports scikit-learn report aggregates alongside confusion matrices so either view can be reconstructed. One-vs-rest TN counts are typically large; practitioners should prioritize TP/FP/FN trends and per-class recall/precision curves when ranking defenses.

Let $\pi_c = n^{(c)}/n_{\text{test}}$ denote the empirical class prior on the test set. A trivial majority classifier that always predicts $\arg\max_c \pi_c$ achieves accuracy $\max_c \pi_c$. Any learned model should be compared against this floor. Because class 0 dominates CIC–UNSW-NB15 augmented labels in our split, high accuracy is necessary but not sufficient for operational usefulness.

Appendix C
Threat Model Variants and Expected Effects

Beyond the single-client label flip studied here, several related adversaries appear in the literature:

- Multi-client label flips: if M of N clients are poisoned at fraction f, FedAvg weights the malicious mass by roughly M/N under equal shards. Dilution weakens unless M is large.
- Feature poisoning: perturbing $\mathbf{x}$ can create stealthier backdoors than label flips alone.
- Model replacement: an attacker scales its update so that $\theta^{t+1} \approx \theta_{\text{malicious}}$ after averaging [4]. Undefended FedAvg is much weaker against this than against naive label noise.
- Trimmed-mean / Krum defenses: robust aggregators [9], [10] can reject outlier updates but may fail under adaptive local model poisoning [13].

Our experiments deliberately isolate the weakest nontrivial poisoning case—label flips on one client—to measure baseline dilution. Stronger attacks would require stronger defenses; we leave that comparison to future controlled studies.

Appendix D
Implementation Hyperparameters

Unless overridden by CLI flags, representative defaults include: Adam learning rate 10^{-3}, batch size 512–1024, stratified test fraction 0.2, and random seed 42. Poison client index is 0 when enabled; poison target class is 0 for target mode. Evaluation always uses the clean held-out test partition. The interactive dashboard at https://fl-research.vercel.app exposes charts for the full 64-run CSV and cached confusion scenarios.

Appendix E
Reproducibility Checklist

To reproduce the study:

1) Obtain CIC–UNSW-NB15 augmented Dataset.csv and Label.csv.
2) Install dependencies from requirements.txt.
3) Run federated_unsw.py with the desired client/round/poison flags, or replay the sweep that produced many_tests_ranked_all.csv.
4) Generate static figures via visualize_many_tests.py and confusion artifacts via visualize_confusion.py.
5) Build the interactive dashboard via visualize_many_tests_interactive.py.
6) Compile this paper with an IEEELaTeX toolchain (tectonic ieee_paper.tex or pdflatex).

Appendix F
Extended Qualitative Summary of Confusion Findings

Across cached confusion scenarios (5-client clean, 5-client 40% target, 3-client 25% target, 1-client clean, 1-client 40% target), we observe:

1) Clean multi-client and poisoned multi-client matrices remain visually similar, consistent with Table II.
2) Rare classes (e.g., indices with tiny support) stay near-zero recall in all settings, indicating a capacity/representation issue rather than a poisoning-only issue.
3) Class 5 frequently receives off-diagonal mass in clean runs, acting as a confusion attractor.
4) Single-client target poisoning uniquely inflates predicted-class 0 mass for several true attack classes, matching the security narrative of missed detections with deceptively stable accuracy.

These qualitative findings reinforce Section VI: federated averaging with multiple clients is a meaningful mitigator for this adversary class, but it does not solve intrinsic minority-class detection difficulty.

Appendix G
On Communication and Systems Cost

Although our simulation ignores network latency, the communication model of Section III still guides deployment planning. For parameter count $p \approx |\theta|$ and N clients, each round costs $\Theta(Np)$ scalars exchanged. If a site can afford only $R_{\max}$ rounds per day, increasing E is the primary lever for progress under IID data. Compression (quantization, sparsification) can reduce payload size but interacts with aggregation semantics and is orthogonal to the poisoning questions studied here. Secure aggregation protocols [7] protect individual updates from the server at additional cryptographic cost; they do not by themselves stop poisoning, because a malicious client can still train on bad labels before encrypting a plausible update.

Appendix H
Ethical and Operational Considerations

Collaborative IDS training must respect data governance: even without sharing raw flows, model updates can leak information through membership or property inference. Differential privacy and secure aggregation are complementary controls. Operationally, any federated detector should be monitored with per-class dashboards; accuracy-only SLOs can hide regressions on rare but high-impact attack types—exactly the failure mode amplified by majority-class targeted poisoning in our single-client experiments.